

Human umbilical cord-derived mesenchymal stem cells induce tissue repair and regeneration in collagen-induced arthritis in rats.

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The immunosuppressive and anti-inflammatory properties of mesenchymal stem/stromal cells (MSCs) have prompted their therapeutic application in several autoimmune diseases, including rheumatoid arthritis (RA). MSCs derived from bone marrow and adipose tissue has earlier been tried with limited success. However, Wharton's jelly present in human umbilical cord is discarded after delivery which makes a rich source of MSCs with least ethical issues. The immunomodulatory properties of human umbilical cord-derived MSCs (UC-MSCs) were evaluated in-vitro on the mononuclear cells from synovial fluid (SF) and peripheral blood of RA patients. The therapeutic potential of UC-MSCs was checked by transplanting the cells in rats with collagen-induced arthritis (CIA). MSCs isolated from Wharton's Jelly significantly suppressed the proliferation and activation of lymphocytes from both peripheral blood as well as SF of RA patients, down-modulated the functions of activated CD4+, CD8+ T-cells, suppressed the secretion of pro-inflammatory cytokines, and induced the expansion of T-regulatory cells. Xenotransplantation of UC-MSCs in CIA rats clearly indicated a sustained impact in terms of slowing down the progression of disease activity and reversal of arthritic processes along with triggering of joint tissue repair mechanisms, which could be observed till 6 weeks post-transplantation. The results from the current study suggest that human umbilical cord is a rich source of MSCs for allotransplantation. The UC-MSCs may be used successfully as a cell-based therapeutic option either in isolation or in conjunction with existing therapeutic drugs not only to relieve the joint inflammation but also regenerate the damaged bone and cartilage tissues in arthritis. The current study highlights the potential use of MSCs as a cell-based therapeutic option for the treatment of inflammatory RA.

Safety study of allogeneic mesenchymal stem cell therapy in animal model.

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Intravenous (IV) infusion of mesenchymal stem cells (MSCs) from nascent tissues like Wharton's Jelly of the umbilical cord is reported to offer therapeutic effects against chronic diseases. However, toxicological data essential for the clinical application of these cells are limited. Thus, this study aimed to determine the safety of IV infusion of Wharton's Jelly derived MSCs (WJ-MSCs) in rats. Fifteen male Sprague-Dawley rats were randomised into the control or treatment group. Each group received an equal volume of saline or WJ-MSC (10×10^6 cell/kg) respectively. The animals were evaluated for physical, biochemical and haematological changes at Week 0, 2, 4, 8 and 12 during the 12-week study. Acute toxicity was performed during Week 2 and sub-chronic toxicity during Week 12. At the end of the study, the relative weight of organs was calculated and histology was performed for lung, liver, spleen and kidney. The findings from physical, serum biochemistry and complete blood count demonstrated no statistically significant differences between groups. However, pathological evaluation reported minor inflammation in the lungs for all groups, but visible healing and resolution of inflammation were observed in the treatment group only. Additionally, the histological images of the treatment group had significantly improved pulmonary structures compared to the control group. In summary, the IV administration of WJ-MSC was safe in the rats. Further studies are needed to determine the long-term safety of the WJ-MSC in both healthy and diseased animal models.

Optimization and scale-up of Wharton's jelly-derived mesenchymal stem cells for clinical applications.

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MSCs are promising candidates for stem cell therapy and regenerative medicine. Umbilical cord is the easiest obtainable biological source of MSCs and the Wharton's jelly of the umbilical cord is a rich source of fetus-derived stem cells. However, the use of MSCs for therapeutic application is based on their subsequent large-scale in vitro expansion. A fast and efficient protocol for generation of large quantities of MSCs is required to meet the clinical demand and biomedical research needs. Here we have optimized conditions for scaling up of WJ-MSCs. Low seeding density along with basic fibroblast growth factor (bFGF) supplementation in the growth medium, which is DMEM-KO, resulted in propagation of more than 1×10^8 cells within a time period of 15 days from a single umbilical cord. The upscaled WJ-MSCs retained their differentiation potential and immunosuppressive capacity. They expressed the typical hMSC surface antigens and the addition of bFGF in the culture medium did not affect the expression levels of HLA-DR and CD 44. A normal karyotype was confirmed in the large-scale expanded WJ-MSCs. Hence, in this study we attempted rapid clinical-scale expansion of WJ-MSCs which would allow these fetus-derived stem cells to be used for various allogeneic cell-based transplantations and tissue engineering.

Immunosuppressive properties of mesenchymal stromal cells derived from amnion, placenta, Wharton's jelly and umbilical cord.

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The role of bone marrow-derived mesenchymal stromal cells (BM-MSC) in preventing the incidence and ameliorating the severity of graft-versus-host disease (GvHD) has recently been reported. However, as the collection of BM-MSC is an invasive procedure, more accessible sources of MSC are desirable. This study aimed to explore the alternative sources of MSC from amnion, placenta, Wharton's jelly and umbilical cord, which are usually discarded. MSC from those tissues were isolated using mechanical dissociation and enzymatic digestion. Their capacity for proliferation and differentiation, and ability to suppress alloreactive T-lymphocytes were studied and compared with those of BM-MSC. MSC derived from amnion, placenta, Wharton's jelly and umbilical cord were similar to BM-MSC regarding the cell morphology, the immunophenotype as well as the differentiation ability. These MSC also elicited a similar degree of immunosuppression, as evidenced by the inhibition of alloreactive T-lymphocytes in the mixed lymphocyte reaction, compared with that of BM-MSC. MSC from umbilical cord and Wharton's jelly had a higher proliferative capacity, whereas those from amnion and placenta had a lower proliferative capacity compared with BM-MSC. The results obtained from this study suggest that MSC from amnion, placenta, Wharton's jelly and umbilical cord can therefore be potentially used for substituting BM-MSC in several therapeutic applications, including the treatment of GvHD.

Umbilical cord revisited: from Wharton's jelly myofibroblasts to mesenchymal stem cells.

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The umbilical cord (UC) is an essential part of the placenta, contributing to foetal development by

ensuring the blood flow between mother and foetus. The UC is formed within the first weeks of gestation by the enclosure of the vessels (one vein and two arteries) into a bulk of mucous connective tissue, named Wharton's jelly (WJ) and lined by the umbilical epithelium. Since their first identification, cells populating WJ were described as unusual fibroblasts (or myofibroblasts). Recent literature data further highlighted the functional interconnection between UC and the resident cells. The UC represents a reservoir of progenitor populations which are collectively grouped into MSCs (mesenchymal stem cells). Such cells have been sourced from each component of the cord, namely the sub-amnion layer, the WJ, the perivascular region, and the vessels. These cells mainly show adherence to the phenotype of adult MSCs (as bone marrow-derived ones) and can differentiate towards mature cell types belonging to all the three germ layers. In addition, cells from human UC are derived from an immunoprivileged organ, namely the placenta: in fact, its development and function depend on the elusion of the maternal immune response towards the semi-allogeneic embryo. This is reflected in the expression of immunomodulatory molecules by UC-derived MSCs. The present paper describes UC structural features and the cell types which can be derived, with a focus on their phenotype and the novel results which boosted the use of UC-derived cells for regenerative medicine applications.

Mesenchymal stem cells derived from Wharton's Jelly of the umbilical cord: biological properties and emerging clinical applications.

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In recent years there seems to be an unbounded interest concerning mesenchymal stem cells (MSCs). This is mainly attributed to their exciting characteristics including long-term ex vivo proliferation, multilineage potential and immunomodulatory properties. In this regard MSCs emerge as attractive candidates for various therapeutic applications. MSCs were originally isolated from the bone marrow (BM) and this population is still considered as the gold standard for MSC applications. Nevertheless the BM has several limitations as source of MSCs, including MSC low frequency in this compartment, the painful isolation procedure and the decline in MSC characteristics with donor's age. Thus, there is accumulating interest in identifying alternative sources for MSCs. To this end MSCs obtained from the Wharton's Jelly (WJ) of umbilical cords (UC) have gained much attention over the last years since they can be easily isolated, without any ethical concerns, from a tissue which is discarded after birth. Furthermore WJ-derived MSCs represent a more primitive population than their adult counterparts, opening new perspectives for cell-based therapies. In this review we will at first give an overview of the biology of WJ-derived UC-MSCs. Then their potential application for the treatment of cancer and immune mediated disorders, such graft versus host disease (GVHD) and systemic lupus erythematosus (SLE) will be discussed, and finally their putative role as feeder layer for ex vivo hematopoietic stem cell (HSC) expansion will be pointed out.

Cartilage Repair in the Knee Using Umbilical Cord Wharton's Jelly-Derived Mesenchymal Stem Cells Embedded Onto Collagen Scaffolding and Implanted Under Dry Arthroscopy.

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Cell-based cartilage repair procedures are becoming more widely available and have shown promising potential to treat a wide range of cartilage lesion types and sizes, particularly in the knee joint. More

recently, techniques have evolved from 2-step techniques that use autologous chondrocyte expansion to 1-step techniques that make use of mesenchymal stem cells (MSCs) embedded onto biocompatible scaffolding. Our 1-step technique has been further developed to provide cell-based cartilage repair using MSCs that have the potential to be used in an off-the-shelf manner, without the need for autologous tissue harvest. Precursor MSCs can be isolated in abundance from the Wharton's jelly of umbilical cord tissue. These cells have been shown to have the desired capacity for proliferation, differentiation, and release of trophic factors that make them an excellent candidate for use in the clinical setting to provide cell-based restoration of hyaline-like cartilage. Although allogeneic in nature, these cells stimulate little or no host immune response and can be stored for long periods while maintaining viability. We present a technique of cartilage repair in the knee using Wharton's jelly-derived MSCs embedded onto scaffolding and implanted in a minimally invasive fashion using dry arthroscopy.

Wharton's jelly mesenchymal stem cells differentiate into retinal progenitor cells.

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Human Wharton's jelly mesenchymal stem cells were isolated from fetal umbilical cord. Cells were cultured in serum-free neural stem cell-conditioned medium or neural stem cell-conditioned medium supplemented with Dkk-1, a Wnt/ β catenin pathway antagonist, and LeftyA, a Nodal signaling pathway antagonist to induce differentiation into retinal progenitor cells. Inverted microscopy showed that after induction, the spindle-shaped or fibroblast-like Wharton's jelly mesenchymal stem cells changed into bulbous cells with numerous processes. Immunofluorescent cytochemical ing and reverse-transcription PCR showed positive expression of retinal progenitor cell markers, Pax6 and Rx, as well as weakly down-regulated nestin expression. These results demonstrate that Wharton's jelly mesenchymal stem cells are capable of differentiating into retinal progenitor cells in vitro.

Wharton's Jelly Mesenchymal Stem Cells Exosomes and Conditioned Media Increased Neutrophil Lifespan and Phagocytosis Capacity.

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Neutrophils are the first cells involved in inflammation and pathogen elimination, but they have a short lifespan. So, strategies for enhancing neutrophil lifespan and activities can be useful in many situations such as patients with immunodeficiencies. Previous researches demonstrated that mesenchymal stem cell (MSC) has anti-apoptotic effects on neutrophils. These multipotent cells have immunomodulatory properties and can be isolated from different tissues. MSCs isolated from Wharton's jelly (WJ-MSCs), a mucosal connective tissue of the umbilical cord, may be better candidates than MSCs obtained from bone marrow or adipose tissue, because WJ-MSCs are younger and protected from damages that are resulted from aging, environmental toxins, and diseases. In addition, they have high proliferative capacity, easier accessibility, and more abundance. It was shown that following in vitro expansion, they are more effective than other sources of MSCs. Cell to cell contact or secretion of soluble factors and exosomes are the main approaches of MSCs in applying their effects. Exosomes and conditioned media (CM) were prepared from WJ-MSCs. Then, neutrophils were isolated and cultured with medium, CM, or exosomes. Then, neutrophil respiratory burst, apoptosis, and phagocytosis capacity were assessed by NBT assay, Annexin V-PI method, and Giemsa staining, respectively. Both treatments improved neutrophil lifespan

and phagocytosis. Only MSC-CM could enhance neutrophil respiratory burst. This research demonstrated that MSC-exosomes and CM have protective effects on neutrophil function and lifespan. It can be concluded that MSC mediators can be responsible factors for protective functions of MSCs on neutrophils.

Umbilical cord-derived Wharton's jelly for regenerative medicine applications in orthopedic surgery: a systematic review protocol.

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Musculoskeletal injuries and conditions affect millions of individuals. These ailments are typically managed by immobilization, physiotherapy, or activity modification. Regenerative medicine has experienced tremendous growth in the past decades, especially in musculoskeletal medicine. Umbilical cord-derived Wharton's jelly is an exciting new option for such therapies. Wharton's jelly is a connective tissue located within the umbilical cord largely composed of mesenchymal stem cells and extracellular matrix components, including collagen, chondroitin sulfate, hyaluronic acid, and sulfated proteoglycans. Wharton's jelly is a promising and applicable biologic source for orthopedic regenerative application. A systematic search will be conducted in PubMed, ScienceDirect, and Google Scholar databases of English, Italian, French, Spanish, and Portuguese language articles published to date. References will be screened and assessed for eligibility by two independent reviewers as per PRISMA guidelines. Articles will be considered without exclusion to sex, activity, or age. Studies will be included if they used culture-expanded, mesenchymal stem/stromal cells of mesenchymal stem cells and/or connective tissue obtained from Wharton's jelly. Studies will be excluded if Wharton's jelly is not the sole experimental examined cell type. Placebos, conventional non-operative therapies including steroid injections, exercise, and NSAIDs will be compared. The study selection process will be performed independently by two reviewers using a reference software. Data synthesis and meta-analysis will be performed separately for clinical and pre-clinical studies. The results will be published in relevant peer-reviewed scientific journals. Investigators will present results at national or international conferences. The protocol was registered on PROSPERO international prospective register of systematic reviews prior to commencement, CRD42020182487 .